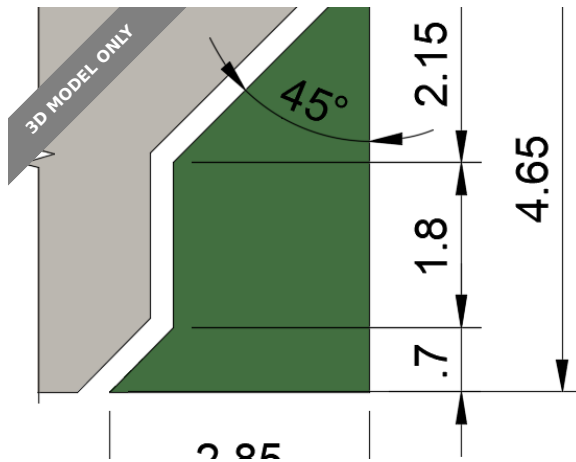




Gridfinity Specification



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[VIEW IN BROWSER](#)

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Summary

Trying to clarify the Gridfinity Specification (read below)

[Hobby & Makers](#) > [Organizers](#)

Tags: [gridfinity](#) [gridfinitybaseplate](#) [gridfinitybin](#)

I am asking for your help. If you see problems with this aggregation, then please leave me comments. If you think all is pretty good, then please click the “like” button or leave me comments as well. Thank you in advance.

This is intended to help myself and others to understand and resolve ambiguities and sometimes conflicting references for the Gridfinity system.

Gridfinity

Gridfinity is a system to help you work more efficiently. Some might classify it as merely an organization methodology, but it is much more. First your tools and materials are “displayed” for easy retrieval. Then think about taking your tools and materials and dynamically, organizing them for optimal usage for any given project. Tools and materials have their “home” but may be pulled out into a desktop grid for use and then

returned to their home, smoothly and easily. And as your needs change you can easily reconfigure how your stuff is stored.

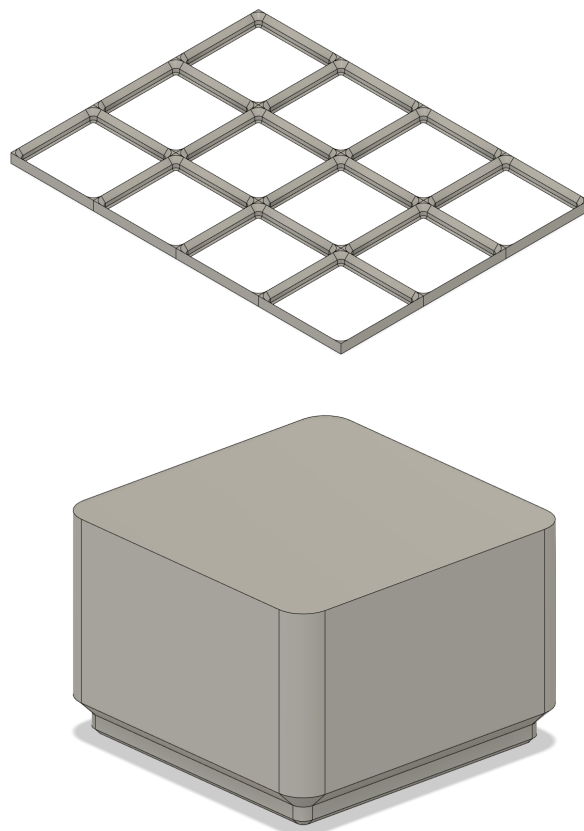
One thing to remember is that gridfinity is a standard and not a rule. Your implementation is YOURS. You can make it whatever you want. However, it is strongly suggested that, when possible, you should follow the standards as doing so will allow you to share other designers' works and allow you to share with others.

Background

Gridfinity is an invention of [Zack Freedman](#), which he introduced in a [YouTube video](#) Apr 12, 2022.

The Magic of 7

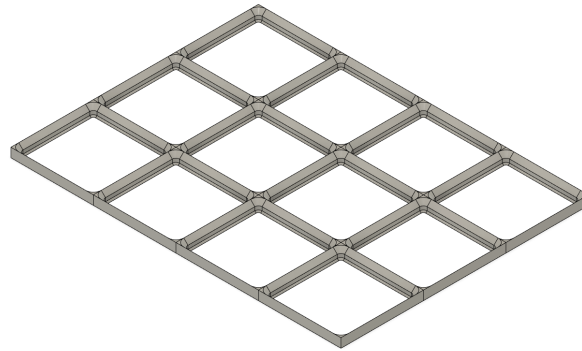
Gridfinity is literally a GRID or matrix of squares. Each square in that grid is 42mm x 42mm in length and width (notice the multiple of 7) and has specially shaped sides to accommodate boxes/bins.



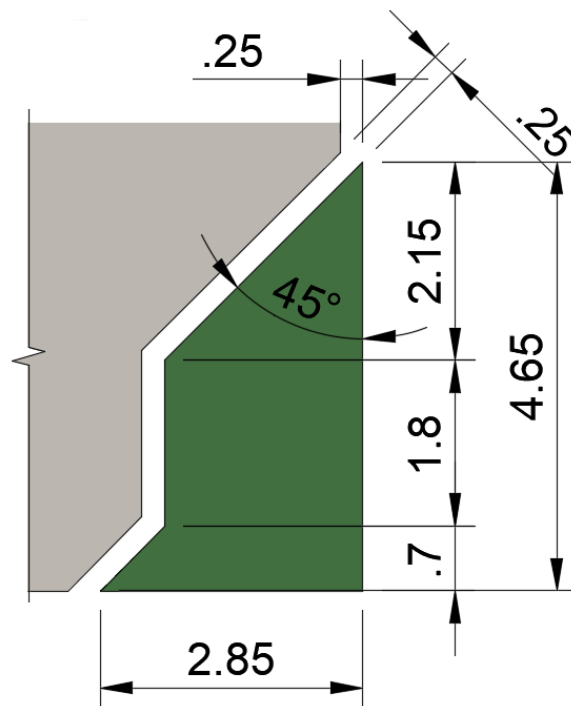
Baseplate

The base grid is known as a “baseplate”. As mentioned earlier, each square is 42mm x 42mm. There are plenty of variations of the baseplate

(more on that later) but all of the variations should follow the 42x42 mm rule.

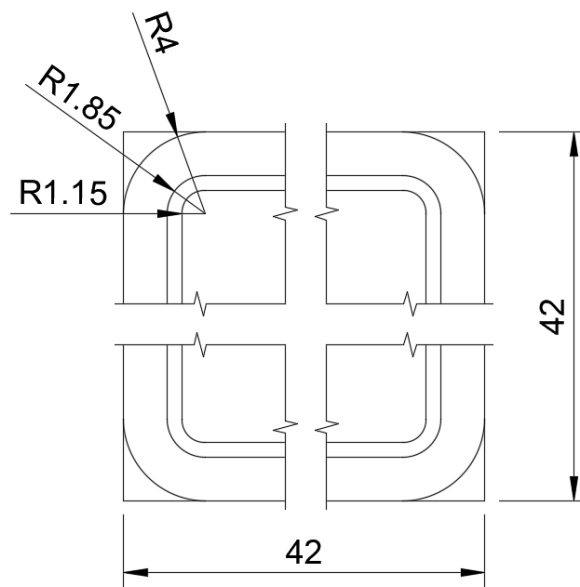


The wall of the basic square is 4.65mm tall and has a slanted stair-stepped profile.

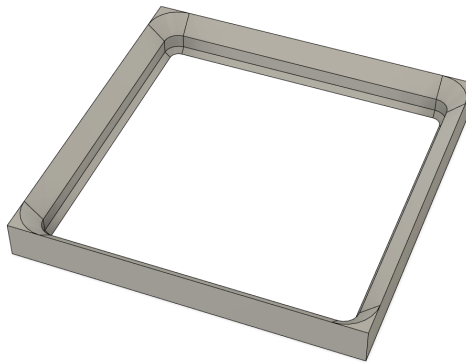


Baseplate
Profile View

The corners have an outside fillet of 4.0mm. The other fillets are based on the widths of the stair-stepped walls of the square. The corners between the squares of the grid are filled solid.



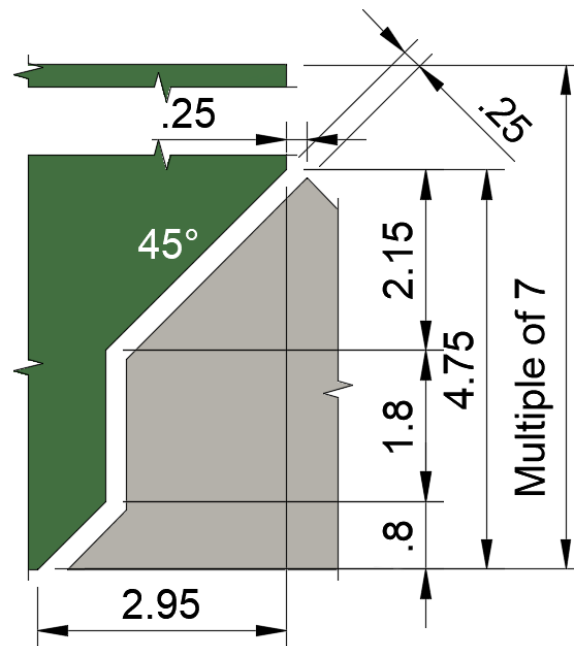
Baseplate
Plan View



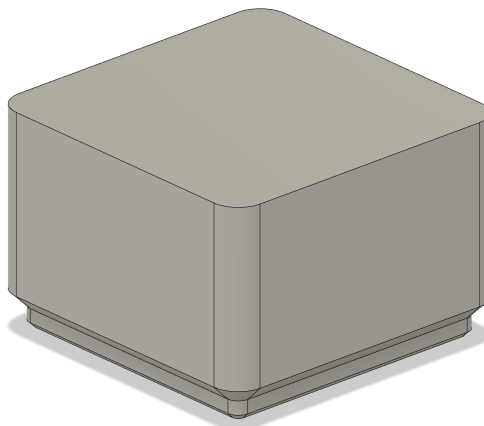
Some implementations have additional features beyond the basic baseplate (more on that later).

Bins/Blocks

The box that is placed within the baseplate is referred to as a “Bin” or “Block” or “Profile” or “Widget” (for our purposes we will use “Bin”). The bottom of the “bin” is shaped to marry with the baseplate. There is a 0.25mm tolerance (gap) between the bin and the containing baseplate square on all sides.

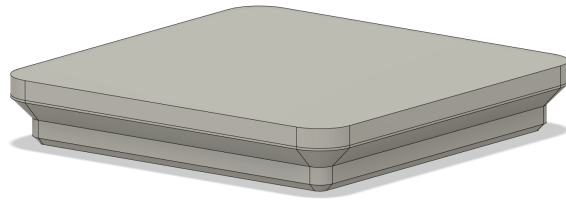


Bin Base
Profile View

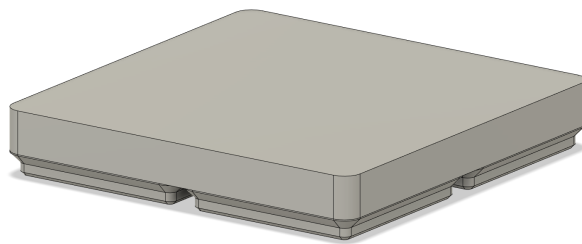


In referring to the “units” of a bin, you specify the integer number of baseplate squares (columns and rows) and the number of integer height units (layers) of 7mm (notice the seven). Bin horizontal dimensions are multiples of 42, but you actually have to subtract 0.5mm (0.25mm + 0.25mm on each side) for the tolerance gap.

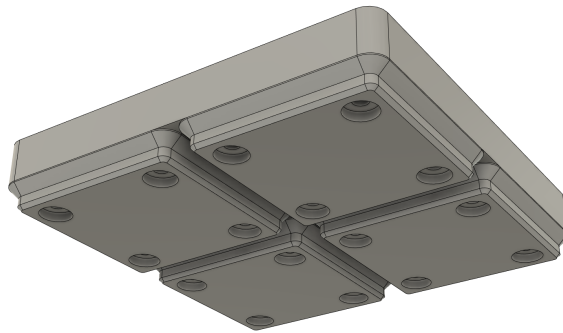
1x1x1 Bin would be 41.5 x 41.5 x 7 mm.



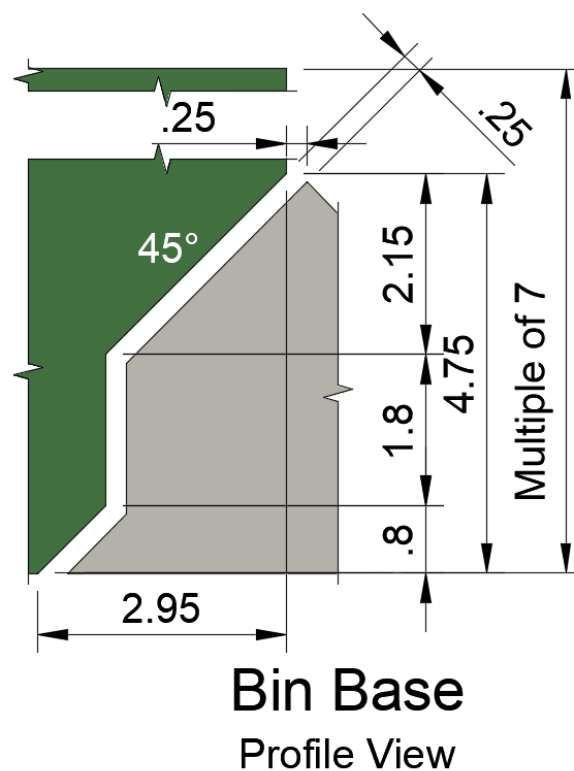
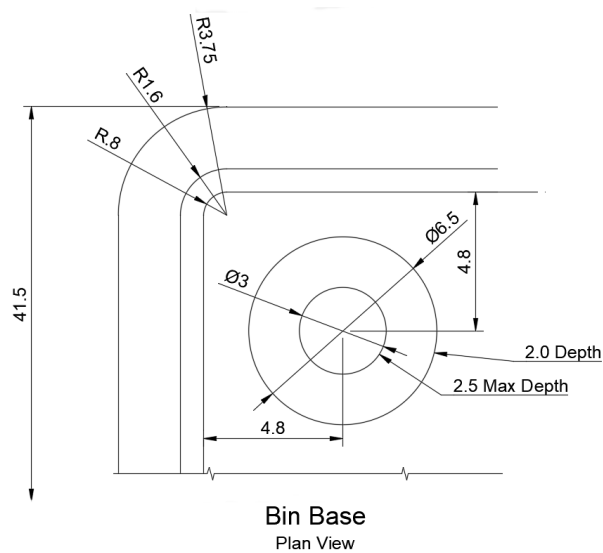
2x2x2 Bin would be 83.5 x 83.5 x 14 mm.



Please notice that bins that span more than one grid square have multiple bin bases that are spaced at 42mm (or 0.5mm between).



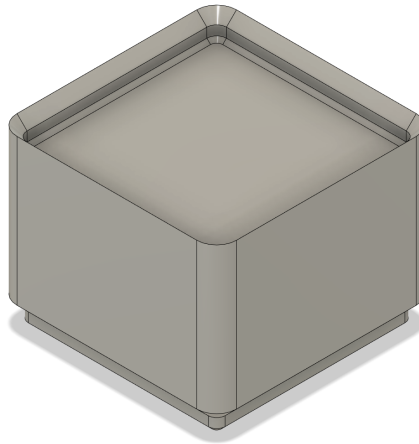
The bins are rounded (filleted) with a radius of 3.75mm (0.25mm less than the baseplate squares).



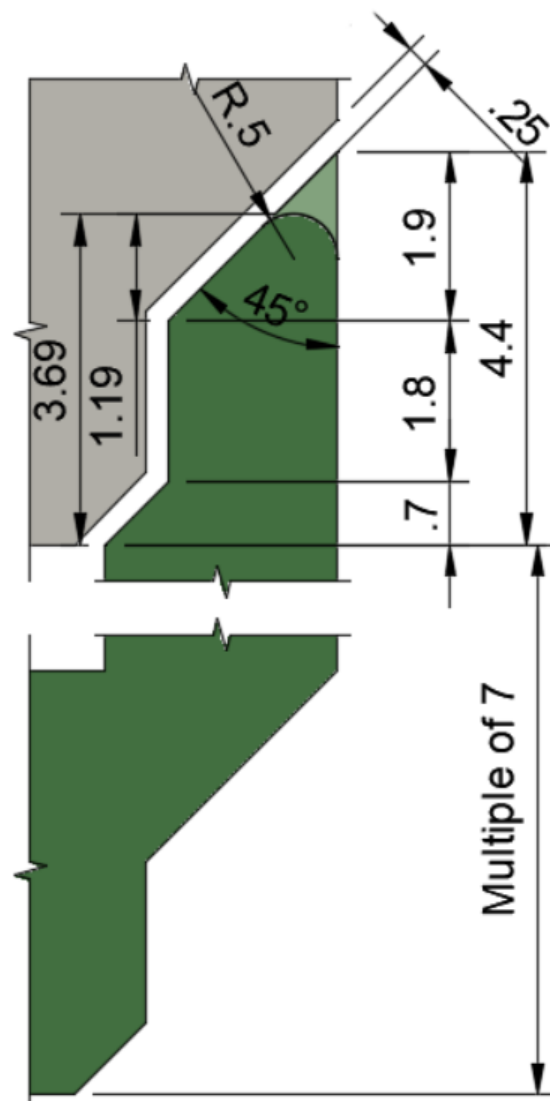
4.75mm of the first layer (7mm) is used up by the bin base, so your first layer has 2.25mm of usable space (probably a little less for strength purposes).

Stacking Lip

An optional “stacking lip” may be added to the top of a bin. This allows one bin to be stacked on top of another.

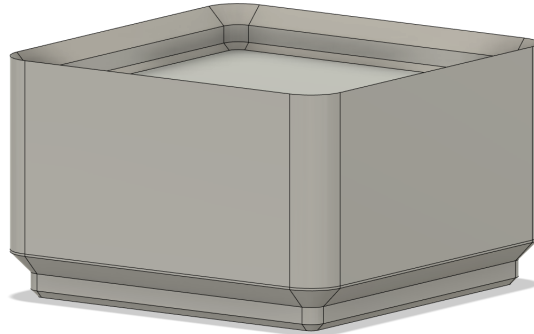


The stacking lip is not included in the integer multiple of 7mm measurement. This is the reason that I earlier indicated “sort-of” for a multiple of 7mm.

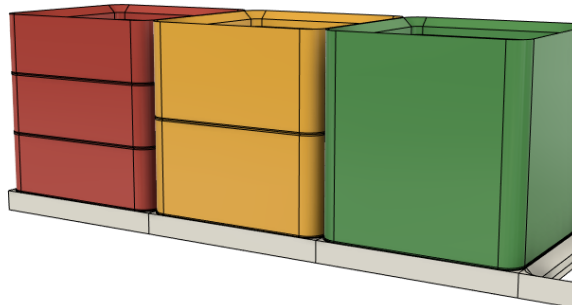


Bin Stacking Lip
Profile View

So, if you have a 1x1x3 bin with a stacking lip, it would be 41.5mm wide x 41.5mm long x 25.4mm tall. Without the lip, the bin would be 21mm tall. If the stacking lip is rounded on the top using 0.5mm the stacking lip would be 3.69mm making the previous height 24.69mm.



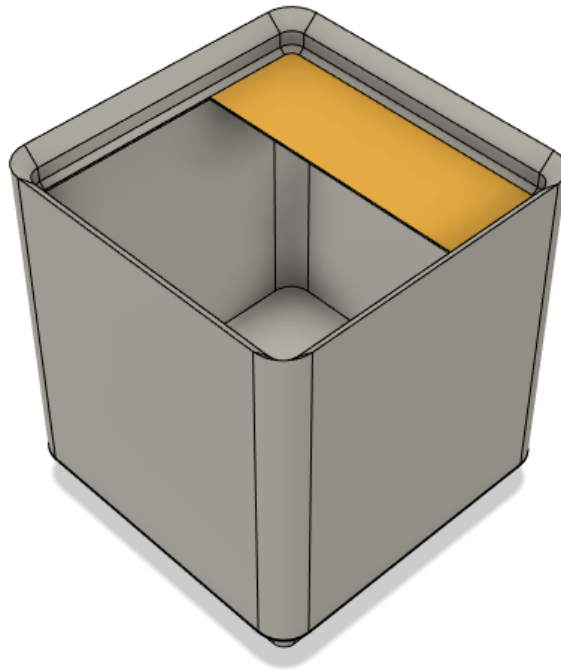
The following example shows how stacking bins of differing heights can add up to be equal. Imagine there are three 1x1x2 bins, two 1x1x3 bins, and one 1x1x6 bin. Notice how they all line up. This is one of the features that Zack Freedman highlights in his [introductory video](#).



If the bin is used in a manner where stacking doesn't make sense, then no lip is needed. As an example, if the bin is used to hold screw drivers, with the shafts sticking straight up, then you can't stack another bin on top, so you don't need a stacking lip.

Bin Labels

Zack provided another very handy feature allowing labels to be added to the bins.



Effectively the label is a shelf at the top of the bin (just below the optional stacking lip). There is no standard regarding the size or shape of the label shelf. The bottom of most labels are sloped at 45 degrees to the wall, so no supports are needed.

The label shelf can also be a handy method of lifting the bin from the baseplate.

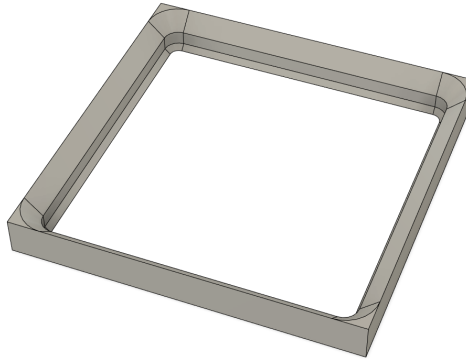
Magnets

Magnets can be optionally added to the bin base.

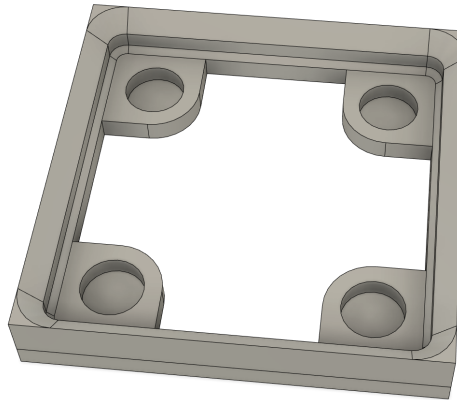
As demonstrated in the [introductory video](#) by Zack Freedman it appears that the original baseplate was targeted for use in a steel drawer. The magnets installed in the bins would help secure them in the drawer. Using a steel surface means that the orientation of the magnets (polarity) does not matter.

Many baseplates now include magnets, so you must be careful regarding a consistent use of the polarity.

As mentioned earlier, the basic baseplates are grids of squares that are 42x42mm.



However, others have extended the implementation to include magnets as in integral part of the baseplate.



Some baseplates have extra weight added to keep them in place on your desk or in the drawer. Think about having a weighted baseplate sitting next to your workspace that contains the bins that are needed for your current task.

Some baseplates are designed to stack on top of a bunch of bins, allowing for a more stable layered approach to organizing the bins. Correspondingly, baseplates can have handles to lift the entire baseplate, with the contained bins, from their storage location.

Zack Freedman also introduced a portable Gridfinity “Toolbox” that contains drawers which, of course, are Gridfinity baseplates.

Others have included holes in the baseplate to allow multiple baseplates to be attached using small nuts and bolts.

Gridfinity Refined has several refinements.

Baseplates have been designed to be attached to other objects including:

- 4040 extrusions or on the top rail of 3D printers

- Pegboards and other wall organizers
- Walls
- Drawers

Bins

Of course, there are dozens and hundreds of bins that are unique in their design as they have been customized to what they are holding. But in addition to those, there is a broad variety of bin features, including:

Vase mode (spiral wall) bins.

Magnets that slide in from the side, as demonstrated by [Gridfinity Refined](#).

Bins that can be attached to the baseplates using [thumbscrews](#).

Bins that lock into the baseplates by having special tabs and slots on the bin and baseplate respectively.

And many, many more...

Gridfinity License

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Model files



baseplate-square.stl



bin-base.stl



baseplate-4x3.stl



bin-w-lip-1x1x3.stl



bin-1x1x1.stl



bin-2x2x2.stl

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